

Efficient Methods for Improving Space Science Education

How can organizations promoting space science optimize their methods for youth space science education to spread knowledge and benefit space policy in the future?

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The space industry began in the late 1950s during the Cold War as another method of non-lethal confrontation between the US and the USSR. NASA has and continues to lead the industry in exploration and innovation, all fueled by government policy (Launius, 2000, p. 23). In the past, space exploration directly fueled US interests, which garnered public support for the industry (Launius, 2000, p. 26). However, reasonings for space exploration have changed throughout the decades, and currently space exploration has benefits that aren't directly pushing towards US goals. Another recent development in the space industry is the growth of the private sector, with companies in the US and across the world positioning themselves to grow in the industry and make a name for themselves (Robinson & Mazzucato, 2019, p. 940). There have also been recent examples of the public and private sectors partnering for the sake of research and exploration, as seen with a recent collaboration of NASA and UC Berkeley (Rausser et al., 2023, p. 4). However, the space industry remains mostly government led in the US and is subject to the same reliance on public opinion as many other government led industries.

Literature Review

Public opinion and space policy

Since the space industry is a government led industry, it is important to recognize that it relies heavily on public opinion. The relationship between public policy and public opinion has been hard to describe in the past for social scientists, with the common phrasing being that there is a linkage between the two (Steinberg, 2011, p. 240). However, after reviewing numerous different historical public policies and the public opinions on each topic, it is shown that public opinion affects public policy a majority of the time and is dependent on many factors (Burstein, 2003, pp. 32-35). One such factor that affects public policy is issue salience, which is the relative

importance an issue has to overall government procedure. The idea of issue salience as a factor in government policy relating to public opinion is more prevalent when understanding the idea of electoral accountability; the idea that elected officials won't stray from their voters' public opinion. Electoral accountability gauges how elected officials rely on their voters' opinions and explains how public opinion is a driving force in government (Shapiro, 2011, pp. 984-986). In the case of space policy, an issue of low salience, public officials are elected for other more prominent issues, and as a result, space policy will take a minimal role in the driving force of electoral accountability.

The biggest driving factor limiting positive public opinion comes from lack of education of the general public in relation to space science and its benefits (Bensaude-Vincent, 2001, p. 107). It is a myth that US spaceflight has been constantly supported by public opinion, even throughout the 1960s at the height of the space race (Whitman Cobb, 2015, p. 13). Data shows that the general public supports spaceflight in pursuit of state goals and achievement rather than just space exploration for the sake of exploration, indicating education of the goals and possibilities for space exploration is lacking (Launius, 2003, p. 169). This lack of education relates to another aspect of public policy in economics, which is a more political concept, but is still limited by the general public's lack of education in the field (Walstad, 1997, p. 202).

Educating America's youth

The most effective way to reach large numbers of children and give them proper education of space science would be through the public education system. There have been many numerous expansion efforts by NASA and other government programs to push space science in k-12 schools to further this education that is sorely lacking in the US (MacLeish et al., 2012, p.

191). However, there are numerous faults in the public education system which would make pushing space science difficult for a large organization such as NASA.

The alternative to formal education through the school system would be informal education through social media. There have been numerous attention grabbing aspects of space science that garner more attention in social media (Hwong et al., 2017, p. 481). Additionally, social media has a larger outreach to education for the less fortunate, while public education is not the same everywhere in the US (Greenhow & Lewin, 2015, p. 8). Connecting this form of informal education and the public school system would be beneficial, but there are still many other methods of learning that are unaccounted for (Hofstein & Rosenfeld, 1996, p. 90).

Connecting youth education to space policy

Our learnings in the previous sections show that youth education would push for improved public policy, pushing for better government policy and therefore space policy. Space exploration is one of the few technological fields that is pioneered by government agencies, and as a result, public opinion has a major effect on space policy changes (Metzger et al., 2013, p. 27). In the US, there is a lack of education about space science and how it benefits our everyday lives. Researching this topic could show how space science knowledge could be spread to the youth of the US in the most efficient way, which could be used by government agencies trying to increase public opinion in regards to space policy. Learning which methods are best for space science education could also have implications towards efficient methods of education in other fields (Shaghghi & Antonakopoulos, 2012, pp. 179-182).

Gap in understanding

Though research has been done in relation to influences on public opinion of space science, there is a lack of information comparing methods of education in relation to space

science. There are many avenues for education towards space science, informal and formal, but finding the best avenue will allow for efficiency in space science education. This leads us to the research question: How can organizations promoting space science optimize their methods for high school space science education to spread knowledge and benefit space policy in the future?

Method

To figure out which method of education is best for educating the youth in space science, the method chosen for research was an experiment. There were also short questions administered before and after the experiment to better control extraneous variables that may affect the results, which will be discussed later.

Participant selection

I aimed to gather 18-21 participants, consisting of 6-7 students for each of the 3 different methods of education present in the experiment. All of the participants were students of a central Texas high school. I gathered participants by offering them incentives such as partial service credits for various honor societies and brownies. The ingredients list for the brownies can be found in Appendix A. All of the participants were sent consent forms and for those under the age of 18, their parents were sent a notification email as seen in Appendix B. All of the participants maintained anonymity through a lettering system. This experiment was given IRB approval.

Experimental procedure

The first step in this experiment was to sit down and ask the opening questions to the participants. Two questions were asked, “rate your current support for space policy on a scale of 1-5” and “what past experiences have you had in regards to that topic and how have they influenced your level of support”. The response to the second question was coded into various categories of influence in space science, those being: family, in-person experience, social media,

school, and curiosity of space. These categories were based on a previous study which attempted to dissect the influences that can change space policy support in young adults (Kalmbach & Ralston, 2021, p. 10). After those answers were recorded, participants were randomly assigned to one of three different methods. The three different methods were a video, a podcast, or reading assessment, all about the benefits of space science. The participants then spent between 15-40 minutes engaging in their respective learning material. Once the participants completed their learning, they were then asked one question, “What were the most interesting aspects of space policy that you learned about and why?”. The response to this question was recorded word for word with no audio being taken. The next step in the experiment was to set up a meeting time about a week later, which took about five minutes of the participant’s time, and was simply a meeting to ask the last question once again. The participant’s response was recorded again word for word. The complete questions and script can be found in Appendix C.

Data analysis

There is one independent variable and one dependent variable in this experiment. The independent variable is the method of education, which will be one of three tools, a video, a podcast, or a reading passage. The Youtube video was created by Mark Rober and is called “Is NASA a waste of money?”. It discusses the benefits of NASA and why the government puts money towards the organization. It is designated as the representation for social media influence due to its nature as a more interactive and relatable experience, as well as being created by a well-known “influencer”. The podcast is called *TechStuff* and the episode title is “Technology we can thank NASA for”. It is another piece of media that describes the technologies NASA has created, and I designated it as a representation of the curiosity factor. The reading passage was published by *The Planetary Society* and is titled “Why space exploration is always worthwhile”.

This passage is similar to the other two pieces of media and describes the benefits of NASA.

This method of education represents the formal education method which is what is being given in science classes through public education. The dependent variable in the study is a qualitative measure on how well the student picked up information. This is based on the student's response to the post-experiment question which was asked directly after the experiment and then again a week later. Both of the responses of question 3 were used to score each participant on their ability to learn about space science. Participants were given a point for each phrase that matched the phrases they said the first time they answered the questions, as well as a point if their second response matched the length of the first response. Thus, the score that they were given was used to compare the three learning methods in the graphs. The first graph compared the average score of participants from the three different groups. The second graph compared the average score of participants from the three different groups with the average prior support each group had, which comes from the pre-experiment questions.

The pre-experiment interviews will have one quantitative and one qualitative measure. The quantitative measure was the participant's prior stance on space policy, and it factored into data analysis when I compared the ability to pick up information about space policy to a participant's prior stances. The qualitative measure that was found was a participant's past influences in space policy, and those were sorted into categories and addressed as limitations of the study. This info was also used as data to help answer the research question. There were two graphs made for this aspect of the data, the first graph being a graph of the number of times each prior influence was present from question 2, and the second graph showed the average prior support from each prior influence based on questions 1 and 2. The average prior support for each method was also used to create the second graph from the experiment portion of the data.

Limitations of the method

Parental, academic, religious, and many other prior influences could impact a person's ability to change or positively view space science. It has been proven that these influences can affect a young adult's stance on space policy (Kalmbach & Ralston, 2021, p. 10). Although this study was specifically targeting young adults, it can be inferred that the same influences that affect the young adult's in this study could be present for our student participants. The pre-interview questions conducted attempted to account for these prior influences, but they will still have an impact on our participant's ability to change their stance.

Another possible source of error in the results could be the varying academic performance levels of the participants. Everyone's brain functions differently, and it is well known that some students are more apt at picking up information than others. During the study, participants were not subject to any questions regarding their general academic performance or any other test which may be able to account for their level of academic understanding. The code used to determine which source of information was the most successful was dependent on the amount of information remembered by each student, allowing for academic performance to not have a big influence on the study.

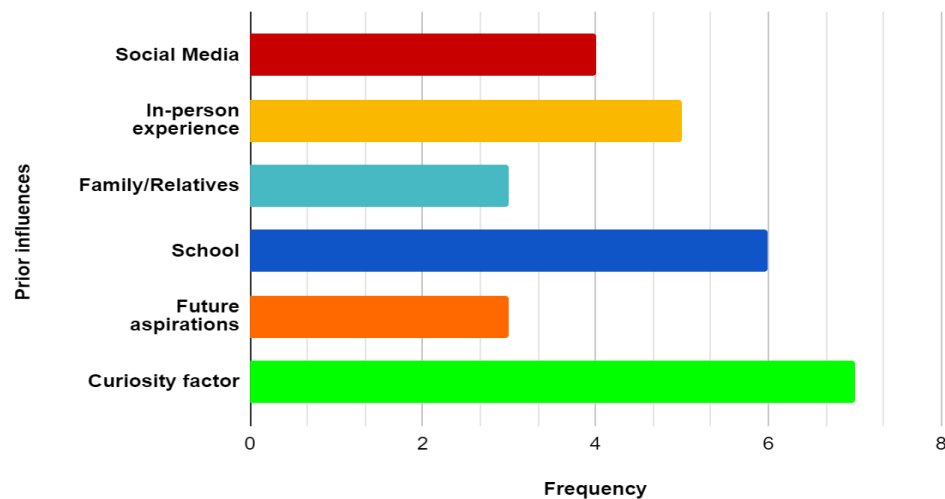
Findings

The purpose of this study is to identify which methods of education will best educate high school students such that that method could benefit future support of space science. My hypothesis going into this study is that social media is the most effective way to educate high schoolers in space science education. Through my research, there were two sections of data which can help us answer this question. The first section will be based on what prior influences have affected students in the past, and whether they have been successful in garnering support

for space policy. The second part was experimenting with different methods of education to see which ones were successful in educating students. Both of these parts will help us come to an overall conclusion regarding how to best educate students and gain support in space science.

Figure 1

Prevalence of prior influences among participants

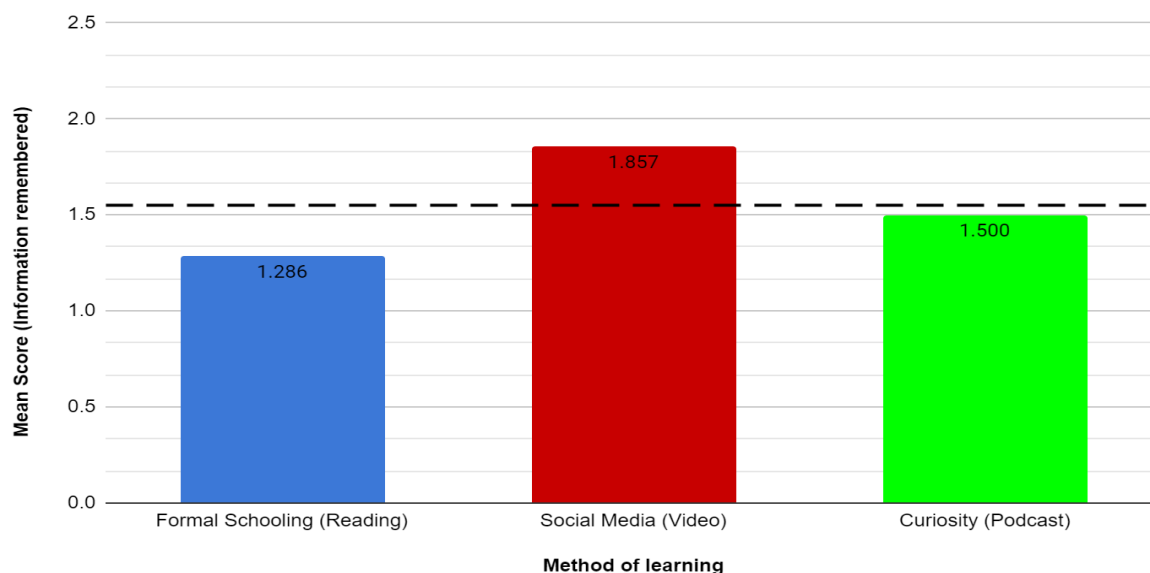


As seen in the graph, the curiosity factor associated with space is one that is most commonly seen in my sample of students. Following closely are the effects of schooling, in-person experiences and social media. The effect of in-person experiences could be heightened for this sample, as it is made up completely of Central Texas high school students, and driving to NASA's Johnson Space Center is only a couple hours away. With the lowest number of answers, both family/relatives and future aspirations have 3 appearances in this sample. Therefore, it can be seen that space science may not be the most popular dinner conversation topic. It can also be seen that in my sample of 20, only 3 students had their sights set on a field relating to space science, which makes sense as many high school students can be undecided on what career they want to pursue.

Each student was allowed to have as many prior influences as they pleased, which is why the total number of each of the bars on the graph doesn't add up to 20, the total number of participants.

Figure 2

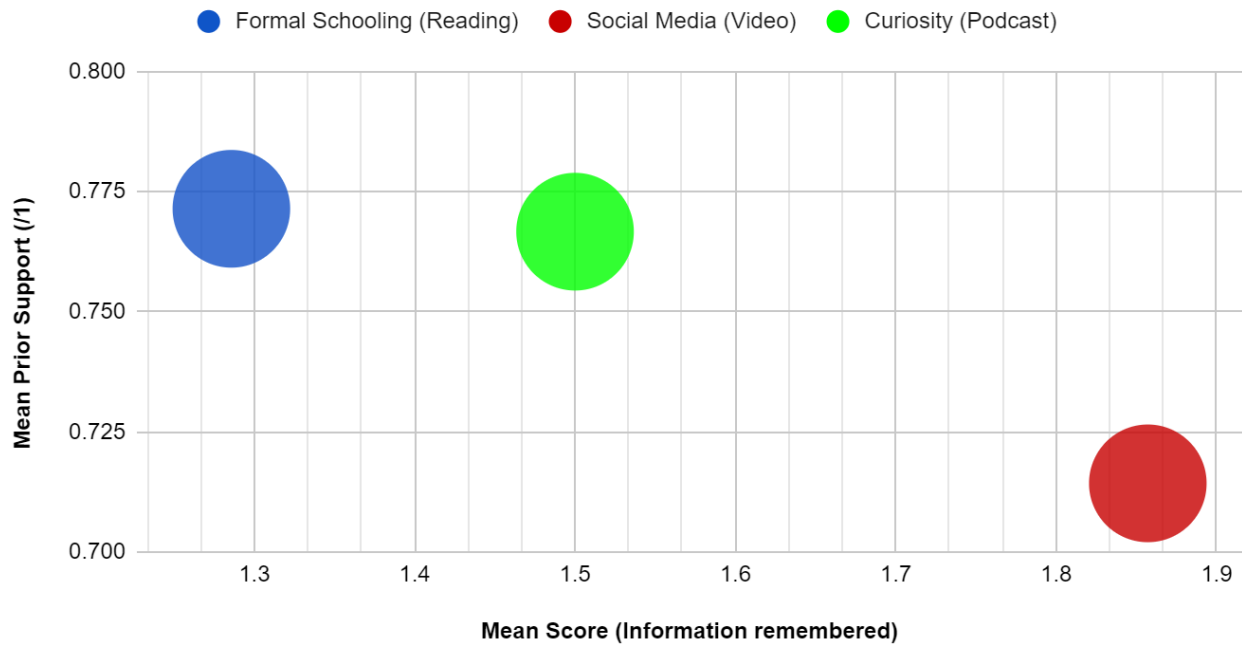
Information remembered for each learning method



This graph moves us on to the experimental portion of the experiment. Each of the 20 participants were scored for their levels of information remembered, and then their averages were taken and put on this graph. The average score among all participants was 1.55, and can be seen by the dashed line across the graph. As seen in the graph, the social media method (video) was the most effective, generating a score almost 20% higher than the average score across all participants. The curiosity factor, represented by the podcast, almost hits the 1.55 average score, with a score of 1.50. The lowest performing score was the formal schooling method, represented by the reading passage, receiving scores just over 20% below the average of 1.55. This information is important to answering the hypothesis, but consideration must be made for the prior influences on the participants.

Figure 3

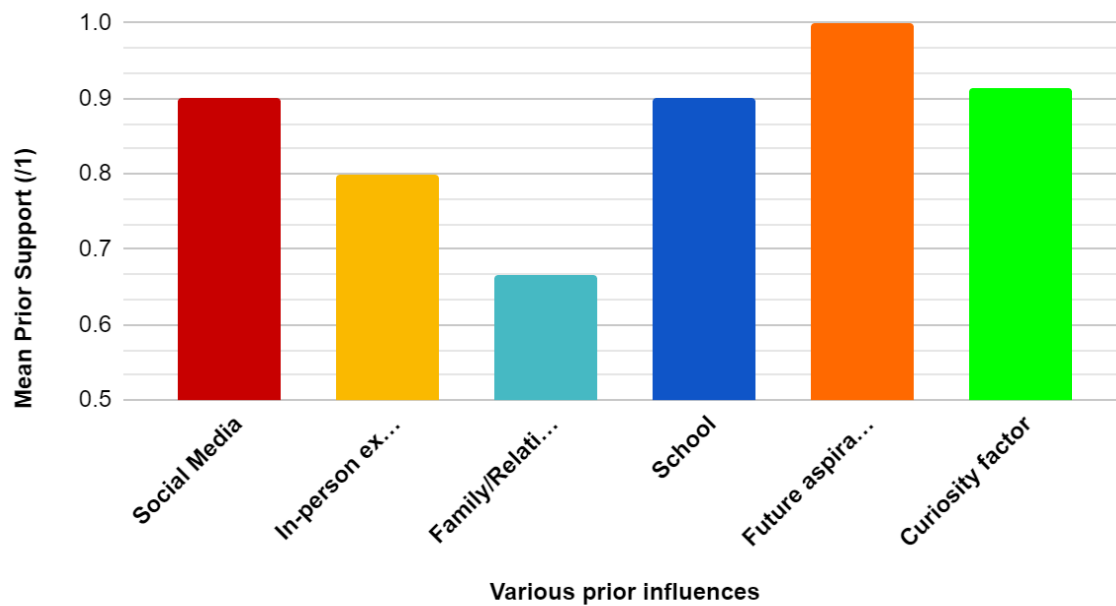
Information remembered compared to prior support level



This graph illustrates the success of the social media method compared to the other two methods. In this graph, it is more beneficial to be in the lower right corner, as that means the participants began with a lower level of support prior to the experiment, but still excelled in remembering information from the video. While it is random which participants receive which method, the lower prior support should theoretically have given the social media method a slight disadvantage compared to the other two methods. While the other two methods, curiosity factor and formal schooling, both started out at similar levels of prior support, the curiosity factor can be seen to be more successful in educating its participants. It is important to also recognize that the scale of the y-axis on this graph is greater, to provide more evidence of the disparity in prior support of the three different methods. Overall, this graph furthers the idea that social media is more effective and supports my hypothesis to the research question.

Figure 4

Support for space policy based on prior influence



This graph brings us back to the short interview questions asked at the beginning of the experiment. As seen in the graph, future aspirations have the highest level of support per person. This makes sense, as many students who are decidedly going into a field related to space science are highly likely to support space policy. The other three highest supported influences are social media, school, and curiosity factor, with social media and school tied for .9 and curiosity factor slightly ahead with a score of .915. In-person experience is also well performing with a score of .8, but yet again that could be influenced by the participants in this study and their proximity to the Johnson Space Center in Houston. With the lowest score, family relationships have a lower level of prior support, which also coincides with it having the lowest frequency of all of the prior influences present. This data is also helpful in answering the research question, and partnered with the experimental scoring portion could give us a significant conclusion.

Analysis

Although the data taken in the experiment was helpful to answering the research question, statistical significance must be taken into account. For Figures 1 and 3, the portion of the data coming from the experimental aspect, statistical analysis was conducted using an ANOVA calculator. The results can be seen in Table 1.

Table 1

ANOVA results comparing each learning method and scores

Source	Degrees of Freedom	Sum of Square	Mean Square	F Statistic	P-value
Groups (between groups)	2	1.1643	0.5821	1.0113	0.3846
Error (within groups)	17	9.7857	0.5756		
Total	19	10.950	0.5763		

This test was used to compare the scores of each group. One of the significant statistics given to us is the F statistic, which is in the 95% region of acceptance, and implies that there are no outstanding differences in means between groups compared to the means in the groups. This is good for the data, but the most important value given is the P-value. Unfortunately, the P-value is .3846, indicating that the data is not statistically significant. Expanding upon that, this P-value indicates that there is too high a chance (38.46%) of rejecting a correct hypothesis, meaning a significant conclusion can't be made based on the data.

Statistical analysis was also conducted based on Figures 2 and 4, regarding the pre-experiment questions and prior influences. A t-test was conducted to compare the relationship between frequency of prior influences and prior support for each prior influence. The two tailed p value was .0002. This value indicates that the data is considered extremely

statistically significant, which means greater significance must be placed on this data to create a conclusion and answer the research question. The results of this aspect of the data are consistent with the results from the less significant aspect of the study, so the differing statistical significance of the two aspects won't cause two different possible answers to the research question. However, before examining the data and crafting an answer to the research question, there are many limitations based around analysis of the data that must be discussed.

Limitations of the data

The most prevalent limitation that appeared in the study after examining the data was the prevalence of in-person experience reported. All of the participants of this study attend a Central Texas High School that is about a 3 hour drive from NASA's Lyndon B. Johnson Space Center in Houston. Along with this, the majority of the high school's student body is made up of the middle or upper class, making traveling to visit the Johnson Space Center a viable option for a majority of the students participating in the study, if not all. Knowing this, the prevalence of in-person experiences must be taken under consideration for feasibility in answering the research question. However, the average prior support, which was a positive .8 (as seen in Figure 4), can be taken into account but still with some caution. Although it is a mean and therefore the prevalence of the in-person experiences isn't significant, the type of in-person experience is important. There must be caution in using the success of in-person experiences in generating prior support for the study as in-person experiences could mean anything from a mini volcano demonstration by a science teacher to seeing a rocket launch. Therefore, the prevalent in-person experience of visiting Johnson Space Center is not as feasible for other high school students in areas not close to any NASA center, especially the midwest and western states (excluding California). Overall, the prevalence and positive support generated from in-person experiences in

this study shouldn't be weighted as importantly as other influences when answering the research question due to the nature of the proximity of student participants to the Johnson Space Center.

Another major aspect of the data that must be discussed is the outstanding prior support for space policy demonstrated by those with future aspirations as a prior influence (shown in Figure 4). Although there were only 3 participants who recorded future aspirations as a prior influence (Figure 2), all of those 3 participants answered a 5 when asked to rate their support for space policy on a scale of 1 to 5. This happens because students who are set on entering a field related to space science are highly likely to support that field and therefore support space policy. In hindsight, it may have been better to clarify what inputs caused someone to go into that field or have those future aspirations, as that may have been more helpful to answering the research question, but the organization of the questions and the study didn't account for that. Through this limitation and the limitations of the results of in-person experience as a prior influence, the other 4 influences in Figure 4 can be seen more clearly as curiosity factor, schooling, and social media tied as the most beneficial prior influence and then family/relatives being by far the least beneficial prior influence.

Answering the question

Having addressed limitations in the data, the research question can be answered and more implications of the study can be evaluated. Before conducting the study, I hypothesized that social media would be the most effective form of space science education for high school students to improve future space policy. This hypothesis has been largely supported by my data, with social media performing exceptionally well in both aspects of my study as seen in Figures 3 and 4. The data from Figure 3 is in extreme favor of social media as the most effective method of education, but it is important to remember that its data isn't statistically significant enough to

make a significant conclusion on. However, the data from Figure 4 also supports social media as one of the most effective methods. This data must be taken accounting for the limitations of the future aspirations and in-person experiences influences, as well as understanding that this data has a high level of statistical significance in accordance with its t-test. In Figures 3 and 4, curiosity factor also performs well, performing better than schooling in Figure 3 and exceeding social media ever so slightly in Figure 4. Thus, it must also be taken into account as a possible answer to the research question. Having discussed all of this, the research question can be answered by saying that social media is the most effective method of education in regards to space science for high schoolers, with other methods inducing the curiosity factor of space being similarly effective in educating high school students.

Coming full circle, this conclusion can bring us back to the lit review and main purpose of this study, which was finding ways to improve public support for space policy. As discussed before, the most important way to improve public support for space policy is to improve education in the field (MacLeish et al., 2012, p. 191). Therefore, it can be concluded that space organizations such as NASA should push for more social media presence to appeal to high schoolers, which will further their education and therefore future support for space policy. With the results of this study, social media was represented by a Youtube video, but social media could be represented by other mediums, although it is unknown based on this study how effective they would be. The other effective method discussed was implementing the curiosity aspect of space exploration. This is the aspect of human's innate desire to explore the unknown, to be curious. Organizations such as NASA could present space and its nature as an endless undiscovered galaxy to push the curiosity aspect of space exploration, therefore furthering future support for space policy among high schoolers.

Conclusion

Overall, this study has shown that social media is one of the most effective ways to educate high school students about space science. It also presents the curiosity factor of space exploration as another way to draw attention to the field. Both of these methods can be used to improve public support for space exploration, therefore garnering support for the industry as a whole.

This study could also have implications for education as a whole outside of just space science. Other fields could see the success of education through social media, and could work to improve their presence on social media. However, the difference between social media of space science and other forms of social media education is unknown right now and could be the subject of future research. This information also is more beneficial to space organizations, which as described in the literature review, makes education of high school students more impactful.

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Appendix A

H-E-B Bakery Two-Bite Brownies

Nutrition Facts

2 brownies (38g)

about 9 servings per container

Serving Size **2 brownies (38g)**

Amount Per Serving

Calories **160**

% Daily Value*

Total Fat 9g **12%**

Saturated Fat 2g **10%**

Trans Fat 0g

Cholesterol 0mg **0%**

Sodium 75mg **3%**

Total Carbohydrate 21g **8%**

Dietary Fiber 1g **4%**

Total Sugars 12g

Includes Added Sugars 12g **24%**

Protein 1g

Vitamin D 0mcg 0%

Calcium 10mg 0%

Iron 1mg 6%

Potassium 90mg 2%

* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.

Ingredients

Sugar, Bleached Enriched Wheat Flour (Wheat Flour, Niacin, Reduced Iron, Thiamine Mononitrate, Riboflavin, Folic Acid), Soybean Oil, Egg Whites, Water, Corn Syrup, Chocolate (Sugar, Chocolate Liquor, Cocoa Butter, Soy Lecithin, Vanilla Extract), Cocoa Powder (Processed With Alkali), Vegetable Oil Shortening (Palm Oil, Canola Oil), Natural And Artificial Flavor, Salt, Whey (Milk), Modified Cornstarch, Leavening Blend (Sodium Acid Pyrophosphate, Sodium Bicarbonate, Cornstarch, Monocalcium Phosphate, Calcium Sulfate), Soy Lecithin, Molasses.

Allergens and safety warnings

Contains: WHEAT, EGG, SOY, MILK. Manufactured in a facility that processes tree nuts.

Appendix B

Space Science Research Interest and Consent

Thank you for your interest in participating in my research project. The project consists of 4 parts, a short interview, a learning experience from one of 3 randomly chosen methods, a test of knowledge after, and then the same test of knowledge a week later. The 3 methods of learning about space science are a watching a YouTube video, listening to a podcast, or reading an article. At all times during the short interviews or experiment, you have the option to opt out.

Not shared

* Indicates required question

If you are interested in my study, please put your contact information here:

Your answer

Do you consent to answering a couple questions about your background for past * experiences with regards to space policy?

☐ Yes
☐ No

Do you consent to learning about space science through one of the methods? *

☐ Yes
☐ No

Do you consent to explaining what you learned directly after learning and then a week after learning? *

☐ Yes
☐ No

Do you consent to spending at least 45 min during this research project? *

☐ Yes
☐ No

If you consent to participating in this research project and answered yes to all of the questions above, please sign your name below. *

Your answer

If you are under 18 years old, please provide your parent's email address here.

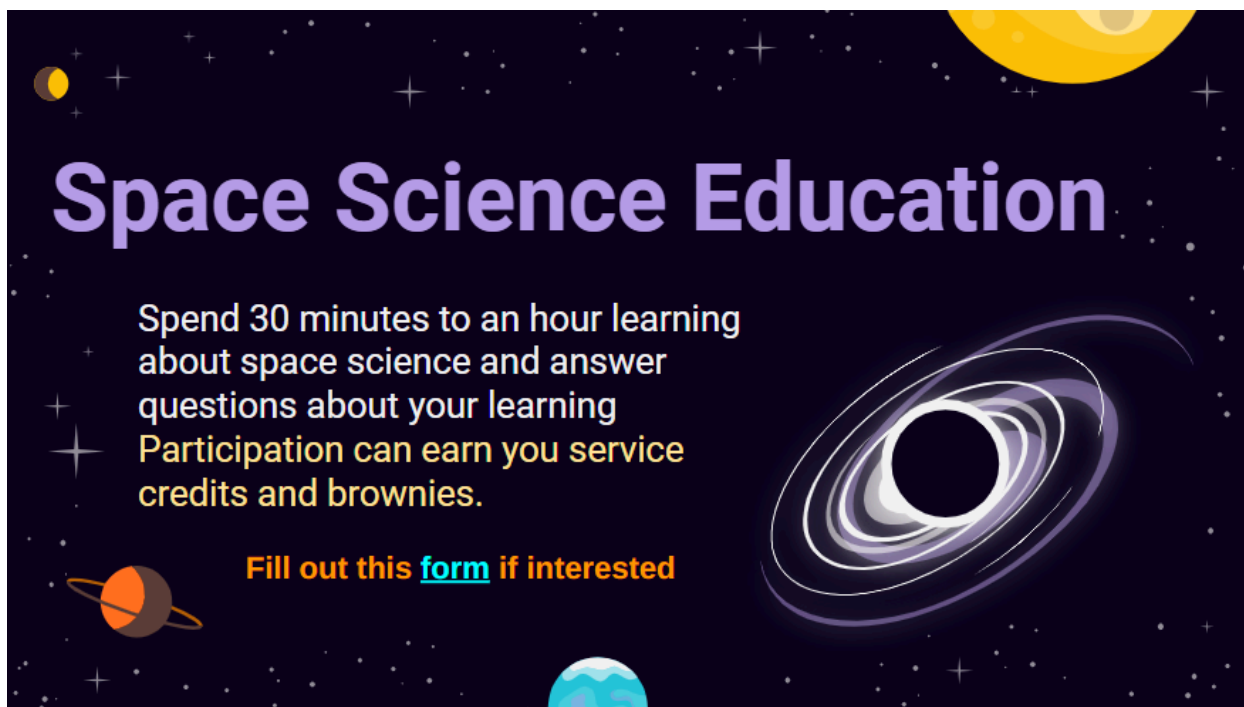
Your answer

Submit Clear form

Never submit passwords through Google Forms.

This form was created inside of Leanderisd.org. [Report Abuse](#)

Google Forms



Appendix C

Questions:

Q1: Rate your current level of support for space policy on a scale of 1-5

Q2: What past experiences have you had in regards to that topic and how have they influenced your level of support?

Q3: What were the most interesting aspects of space policy that you learned about and why?

Script:

“Thank you for participating in my research project. For the first part, I’m going to ask you a few short questions regarding space policy. You can opt out of any of the questions if you would like but none of them are objectively personal. You will maintain anonymity for all purposes this research may be used for”

Ask Q1 and Q2

“Alright, now you will be randomly assigned to either watch a Youtube video, listen to a podcast, or read an article on space policy. Please let me know when you have finished watching/listening/reading the video/podcast/article. ”

Give the participant access to the video/podcast/article

Wait for the participant to complete the task

“Now I’m going to ask you a single question”

Ask Q3

“The last thing for us to do is schedule a time next week for us to meet for about 5 minutes”

Schedule a time and put in calendar

“Thanks for participating so far, you are welcome to have a brownie and I’ll see you at (agreed on day and time)”

One week later at agreed on time

“Today all you will be required to do is answer one question.”

Ask Q3

“That is all the data I need for my project, thank you for participating. All of your responses will remain anonymous, and you can have a brownie if you would like.”

End